

Week 5: Visualisation with matplotlib

AI Frameworks

Today's learning goals

By the end of this lecture you should be able to:

1. Explain why we plot **before** we model
2. Build histograms, bar charts, scatter plots, and simple line plots
3. Add clear labels, titles, and legends—without clutter
4. Read a chart carefully (scale, outliers, misleading axes)

Agenda (80 minutes)

1. Why plot? (EDA mindset)
2. matplotlib basics
3. Core chart types
4. Labels and good defaults
5. Reading charts carefully
6. A peek at seaborn

Plot before you model

Charts help you:

- See distributions and typical values
- Spot outliers and data-entry errors
- Notice relationships (e.g. size vs price)
- Communicate findings to humans

A model on misunderstood data is confident nonsense.

Exploratory Data Analysis

Look at the data with summaries *and* pictures until the story is clearer.

Import pattern

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 df = pd.read_csv("data.csv")
5
6 plt.hist(df["score"])
7 plt.xlabel("Score")
8 plt.ylabel("Count")
9 plt.title("Score distribution")
10 plt.show()
```

- Create a plot with a plotting function
- Decorate: labels, title, legend
- `plt.show()` to display (especially in scripts)

In Jupyter, plots often appear under the cell automatically.

Which chart when?

Goal	Try
Distribution of one number	Histogram
Compare categories	Bar chart
Relationship of two numbers	Scatter
Change over time / order	Line

Histogram

```
1 plt.hist(df["score"], bins=10, edgecolor="black")
2 plt.xlabel("Score")
3 plt.ylabel("Count")
4 plt.title("How scores are spread")
5 plt.show()
```

Bar chart

```
1 counts = df["genre"].value_counts()
2 plt.bar(counts.index.astype(str), counts.values)
3 plt.xticks(rotation=45, ha="right")
4 plt.ylabel("Count")
5 plt.title("Titles per genre")
6 plt.tight_layout()
7 plt.show()
```

Scatter plot

```
1 plt.scatter(df["size"], df["price"], alpha=0.7)
2 plt.xlabel("Size")
3 plt.ylabel("Price")
4 plt.title("Size vs price")
5 plt.show()
```

alpha helps when points overlap.

Line plot

```
1 # example: temperature by day index
2 plt.plot(df["day"], df["temp"])
3 plt.xlabel("Day")
4 plt.ylabel("Temp")
5 plt.title("Temperature over days")
6 plt.show()
```

Quiz 1 — Chart choice

Quiz

You want to see whether taller people tend to have higher shoe size (two numeric columns). Best first chart?

1. Scatter plot
2. Pie chart of names
3. Only `df.shape`
4. A 3D rotating logo

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer — Quiz 1

Answer

1. Scatter plot — two numeric variables on x and y .

Minimum good manners

- Axis labels with units when relevant
- A short, honest title
- Legend only if multiple series need names

Avoid: neon everything, 3D bars, chartjunk.

Not overdecorating

- One message per figure when learning
- Prefer clarity over “impressive” effects
- If ink does not help understanding, remove it

Read carefully

Ask:

- What do the axes mean? What is the scale?
- Are there outliers that dominate the picture?
- Does a truncated y -axis exaggerate a tiny change?
- Is correlation looking like causation? (usually no)

Outliers

- Points far from the rest may be errors—or real extremes
- Do not delete silently; investigate
- Note them in your write-up

Common mistake

Fitting a story to noise in a tiny sample.

Quiz 2 — Reading charts (true/false)

Quiz

True or false?

A scatter plot that slopes upward proves that x *causes* y .

1. True
2. False

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

False. Association is not causation—many third factors can explain a trend.

seaborn: nicer defaults

```
1 import seaborn as sns
2
3 sns.histplot(data=df, x="score")
4 plt.show()
5
6 sns.scatterplot(data=df, x="size", y="price", hue="city")
7 plt.show()
```

Optional helper this week—matplotlib alone is enough for the practice.

Quiz 3 — Why plot first?

Quiz

A main reason to plot before modelling is:

1. Plots replace the need for any data
2. To spot structure, errors, and relationships early
3. Because models cannot use numbers
4. To avoid using pandas

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

2. Visual checks catch issues and suggest what to model.

Key takeaways

1. Plot as part of EDA—before fancy models
2. Match chart type to the question
3. Label clearly; avoid decoration noise
4. Read scales and outliers sceptically

Next up

Practice: three required plots + a short “story” paragraph.

Week 6: feature types, encoding, and scaling intuition.